IGCSE · Cambridge (CIE) · Maths

2 hours 39 questions

Target Test

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Total Marks /123

1 The fares for a train journey are shown in the table below.

From London to Marseille	Standard fare	Premier fare
Adult	\$84	\$140
Child	\$60	\$96

For one journey from London to Marseille, the ratio

number of adults : number of children = 11 : 2.

There were 220 adults in total on this journey.
All of the children and 70% of the adults paid the standard fare. The remaining adults paid the premier fare.
Calculate the total of the fares paid by the adults and the children.

\$ ....

(5 marks)

2 Write down a prime number between 20 and 30.

(1 mark)

3 Dhanu has a model railway.

He has a train that consists of a locomotive and 4 coaches. The mass of the locomotive is 87 g and the mass of each coach is 52 g.

i) Work out the total mass of the train.

.... g [2]

ii) Work out the mass of the locomotive as a percentage of the total mass of the train.

.... % [1]

(3 marks)

4 Calculate the value of $(2.3 \times 10^{-3}) + (6.8 \times 10^{-4})$.
Give your answer in standard form.

(1 mark)

5 Work out $\frac{5}{6} \div 1\frac{1}{3}$.

You must show all your working and give your answer as a fraction in its simplest form.

(3 marks)

6 (a) Rationalise the denominator of $\frac{12}{\sqrt{3}}$.

(2 marks)

(b) Work out the value of $\left(\sqrt{2} + \sqrt{8}\right)^2$.

(2 marks)

7 The length of a roll of ribbon is 30 metres, correct to the nearest half-metre.

A piece of length 5.8 metres, correct to the nearest 10 centimetres, is cut from the roll.

Work out the maximum possible length of ribbon left on the roll.

..................metres

(3 marks)

8 Tomas has two cars.

i) The value, today, of one car is $21 000.
The value of this car **decreases** exponentially by 18% each year.

Calculate the value of this car after 5 years. Give your answer correct to the nearest hundred dollars.

$ .... [3]

ii) The value, today, of the other car is $15 000.
The value of this car **increases** exponentially by x% each year. After 12 years the value of the car will be $42 190.

Calculate the value of x.

x = ... [3]

(6 marks)

9 Liz buys packets of coloured buttons.

There are 8 red buttons in each packet of red buttons.
There are 6 silver buttons in each packet of silver buttons.
There are 5 gold buttons in each packet of gold buttons.

Liz buys equal numbers of red buttons, silver buttons and gold buttons.

How many packets of each colour of buttons did Liz buy?

.... packets of red buttons

.... packets of silver buttons

.... packets of gold buttons

(3 marks)

10 (a) Show that $0.\dot{1}\dot{9}$ can be written as $\frac{19}{99}$.

(3 marks)

(b) Explain how $\frac{19}{99} = 0.\dot{1}\dot{9}$ can be used to find $\frac{19}{990}$ as a decimal and write down its value.

$$\frac{19}{990} = \ldots\ldots\ldots\ldots\ldots$$

(2 marks)

11 (a) Adele, Barbara and Collette share \$680 in the ratio 9 : 7 : 4.

Show that Adele receives \$306.

(2 marks)

(b) Calculate the amount that Barbara and Collette each receives.

Barbara \$ ....

Collette \$ ....

(3 marks)

12 A bag contains 54 red marbles and some blue marbles.
36% of the marbles in the bag are red.

Find the number of blue marbles in the bag.

(2 marks)

13 Collette has \$136 and spends half of her money on clothes and $\frac{1}{5}$ of her money on books.

Calculate the amount she has left.

\$ ....

(3 marks)

14 In 2003, Jerry bought a house.

In 2007, Jerry sold the house to Mia.
He made a profit of 20%

In 2012, Mia sold the house for £162 000
She made a loss of 10%

Work out how much Jerry paid for the house in 2003

(3 marks)

15 (a) Write 640 000 000 in standard form.

(1 mark)

(b) Work out $(3 \times 10^7) \div (6 \times 10^4)$
Give your answer in standard form.

(2 marks)

16 1, 2, 3, 5 and 7 are all common factors of two numbers.
Write down the digit that the two numbers must end in.

(1 mark)

17 A box contains red small balls, red large balls, blue small balls, and blue large balls.

The ratios of the balls are

$$\text{red balls} : \text{blue balls} = 5:6$$
$$\text{small balls} : \text{large balls} = 3:8$$

Express the total number of red balls as a fraction of the total number of large balls.

(3 marks)

18 (a) $$T = \sqrt{\frac{w}{d^3}}$$

$$w = 5.6 \times 10^{-5}$$

$$d = 1.4 \times 10^{-4}$$

Work out the value of *T*.
Give your answer in standard form correct to 3 significant figures.

(2 marks)

(b) *w* is increased by 10% *d* is increased by 5%

Lottie says,

"The value of *T* will increase because both *w* and *d* are increased."

Lottie is wrong.
Explain why.

(2 marks)

19 Ollie invests $200 at a rate of 0.0035% per day compound interest.

Calculate the value of Ollie's investment at the end of 1 year. [1 year = 365 days.]

$ ....

(2 marks)

20 Work out $1\frac{1}{7} \times 2\frac{1}{10}$.

You must show all your working and give your answer as a mixed number in its simplest form.

(3 marks)

21 Show that $\frac{4 + \sqrt{8}}{\sqrt{2} - 1}$ can be written in the form $a + b\sqrt{2}$, where a and b are integers.

Show each stage of your working clearly and give the value of a and the value of b.

(3 marks)

22 The length of the side of a square is 12 cm, correct to the nearest centimetre.

Calculate the upper bound for the perimeter of the square.

.... cm

(2 marks)

23 Work out $\frac{2}{3} + \frac{1}{4} \times \frac{2}{3}$.

Write down all the steps of your working and give your answer as a fraction in its simplest form.

(4 marks)

24 Show that $\dfrac{\sqrt{150}-\sqrt{6}}{\sqrt{2}\times\sqrt{3}}$ simplifies to an integer.

(3 marks)

25 Bill buys and sells laptops.

Last month Bill bought 50 laptops. He paid £400 for each laptop.

He sold

40 of these laptops at a profit of 30% on each laptop

10 of these laptops at a profit of 15% on each laptop

Bill's target last month was to sell all 50 laptops for a total of at least £25 000

Did Bill reach this target?

(5 marks)

26 Work out 12.5×9.2.

(3 marks)

27 Show that $\dfrac{6-\sqrt{8}}{\sqrt{2}-1}$ can be written in the form $a + b\sqrt{2}$ where a and b are integers.

(3 marks)

28 Calculate $(3 \times 10^{-3})^3$.

Give your answer in standard form.

(1 mark)

29 Show that $\dfrac{\sqrt{12}}{\sqrt{3+2}}$

can be written in the form $a\sqrt{b}$ where a is a simplified fraction and b is an integer.

(2 marks)

30 (a) The table shows the surface area of some countries and their estimated population in 2017.

Country	Surface area (km^2)	Estimated population in 2017
Brunei	5.77×10^3	433 100
China	9.60×10^6	1 388 000 000
France	6.41×10^5	67 000 000
Maldives	3.00×10^2	374 600

Find the total surface area of Brunei and the Maldives.

.... km^2

(1 mark)

(b) The ratio $\text{surface area of the Maldives : surface area of China}$ can be written in the form $1 : n$.

Find the value of n.

$n =$ ....

(2 marks)

(c) Find the surface area of France as a percentage of the surface area of China.

.... %

(2 marks)

(d) Find the population density of the Maldives.

[Population density = population ÷ surface area]

..............................people/km^2

(2 marks)

31 Simplify

$$\sqrt{32} + \sqrt{98}$$

(2 marks)

32 $\sqrt{5}\left(\sqrt{8} + \sqrt{18}\right)$ can be written in the form $a\sqrt{10}$ where a is an integer.

Find the value of a.

(3 marks)

33 The number of passengers on a train increases from 63 to 77.

Calculate the percentage increase.

....%

(3 marks)

34 Rationalise the denominator of $\dfrac{4}{7-\sqrt{5}}$

Show each stage of your working.

Give your answer in the form $a + b\sqrt{5}$ where a and b are fractions in their simplest forms.

(3 marks)

35 i) 29 parcels each have a value of \$68. By writing each of these numbers correct to 1 significant figure, find an estimate for the total value of these 29 parcels.

\$ .... [1]

ii) Without doing any calculation, complete this statement.

The actual total value of these 29 parcels is less than the answer to **part (i)**

because .... [1]

(2 marks)

36 Find the exact value of $8^{\frac{2}{3}} \times 49^{-\frac{1}{2}}$.

(2 marks)

37 Mohsin has 720 apple trees on his farm.

Each apple tree produces 16 boxes of apples each year. One box contains 18 kg of apples.

Calculate the total mass of apples produced by the 720 trees in one year. Give your answer in standard form.

.... kg

(3 marks)

38
$$a = \sqrt{8} + 2$$
$$b = \sqrt{8} - 2$$
$$T = a^2 - b^2$$

Work out the value of T.

Give your answer in the form $c\sqrt{2}$ where c is an integer.

(4 marks)

39 (a) Ian invested an amount of money at 3% per annum compound interest.

At the end of 2 years the value of the investment was £2652.25

Work out the amount of money Ian invested.

(3 marks)

(b) Noah has an amount of money to invest for five years.

Saver Account	Investment Account
4% per annum compound interest.	21% interest paid at the end of 5 years.

Noah wants to get the most interest possible.

Which account is best?
You must show how you got your answer.

(2 marks)